Completeness of the Milky Way globular-cluster catalogue from the phases of cluster orbits

Draft prepared with Claude Code for V. Belokurov

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Abstract

We estimate how the completeness of the Milky Way globular-cluster (GC) catalogue depends on absolute magnitude and heliocentric distance, using the 165 GCs with full six-dimensional phase-space data. The estimator conditions on each cluster's own orbit: if clusters are phase-mixed, the present position of a detected cluster samples its orbit with probability proportional to the completeness C at that point. Maximising the resulting likelihood needs no model of the intrinsic GC density. On mock catalogues with known selection it recovers completeness without bias and with calibrated intervals (recovered/true = 0.97–0.99, 68% coverage 0.67–0.71). A fixed-target orbit-superposition estimator in the spirit of Khoperskov et al. (2025) is biased high by $\approx 25\%$ on the same mocks. Applied to the data with C a function of extinction-free apparent magnitude V, completeness falls from 1 at $V < 6$ (assumed) to 0.14 at $V > 11$. Weighting each cluster by the inverse of its orbit-averaged completeness turns 165 observed GCs into 483 (68%: 335–765). Adding extinction along every orbit from 3D dust maps (DECaPS, Bayestar19, Marshall, with SFD beyond their reach) improves the fit by $\Delta\mathrm{AIC} \approx 8$ and *lowers* the corrected total: without extinction the fitted $C(V)$ absorbs the dust losses of the inner Galaxy and applies them in every direction, inflating the weights of distant halo clusters. A continuous $C(V)$ whose faint tail keeps falling fits better still and removes a factor-of-two undercount of faint clusters in mocks. Letting completeness differ between the inner Galaxy ($|l| < 30°$, $|b| < 15°$) and the rest of the sky recovers such an effect in mocks; on the data it is preferred without extinction ($\Delta\mathrm{AIC} = -4.2$) but not once the maps are included, where two- or three-region variants change the total by -12 to $+12\%$. Our final estimate (continuous $C(V)$, maps, own-orbit weights) is $N_{\mathrm{GC}} \approx 430^{+215}_{-125}$ (68%, sampling and quantified systematics), of which about 210 (125–325) are fainter than $M_V = -6.5$; three $1/V_{\max}$-type weightings over orbit populations give 393–432 (faint 177–208). In mocks matched to the observed sample the bootstrap intervals are calibrated (68% coverage 0.63–0.79) and no weighting is significantly better than another. The data favour at most a mild increase of orbit size towards fainter clusters. The dominant uncertainties are systematic: the extinction treatment (a factor of 1.7), the assumption that the brightest clusters are complete, and the Galactic potential.

1 How the method works, in brief

The idea. If a cluster is caught in only a small fraction of the places its orbit takes it, it must have siblings that sit in the uncaught parts of their orbits right now. We count them.

Ingredients. For each catalogued cluster we integrate its orbit for 5 Gyr and record it at 500 moments. At each moment we compute how bright it would look from the Sun, $V = M_V + 5\log_{10}(d/10\,\mathrm{pc}) + A_V$. This gives the cluster's *time fractions*: how long it spends looking like each value of V. The catalogue adds one fact per cluster, the V it has today.

Logic. If every cluster were detected, today would be a random moment of each orbit, and clusters would be caught at each V in proportion to the time they spend there. If clusters looking faint are harder to detect, we catch them preferentially in their bright phases. The mismatch between where clusters are caught and where they spend their time measures how detection falls with V.

A worked example. Take two brightness bins, bright (B) and faint (F), and one unknown, $r = C_F/C_B$: how much less often a cluster is detected while it looks faint than while it looks bright ($r = 1$: no selection; $r = 0$: never detected when faint). Three clusters:

Cluster	Time looking bright	Time looking faint	Seen today
1	50%	50%	B
2	20%	80%	B
3	10%	90%	F

The chance of catching a cluster in a bin is its time there times the detection probability there, divided by the same sum over both bins. For cluster 1, caught bright, this is $0.5\,C_B/(0.5\,C_B + 0.5\,C_F) = 1/(1+r)$: C_B cancels, so the positions measure only the ratio r. For clusters 2 and 3 the chances are $1/(1+4r)$ and $9r/(1+9r)$. The clusters are independent, so the probability of seeing all three where they are is the product, the likelihood

$$L(r) = \frac{1}{1+r} \times \frac{1}{1+4r} \times \frac{9r}{1+9r}. \tag{1}$$

It is 0.090 at $r = 1$, 0.277 at $r = 0.25$, 0.314 at $r = 0.13$ and 0.246 at $r = 0.05$. Its maximum is at $r \approx 0.13$. Clusters 1 and 2 spend most of their time faint but are caught bright, which rules out weak selection. Cluster 3 is caught faint, which rules out $r \approx 0$. The fit is the compromise.

The anchor. The likelihood fixes only ratios. Assuming bright-looking clusters are all detected, $C_B = 1$, gives $C_F = 0.13$. This is set once, for the whole function, not per cluster.

The weights. Each cluster's overall detection chance q is C averaged over its own time fractions: $q_1 = 0.5 + 0.5 \times 0.13 = 0.565$, $q_2 = 0.2 + 0.8 \times 0.13 = 0.304$, $q_3 = 0.1 + 0.9 \times 0.13 = 0.217$. Each stands for $1/q$ clusters: $1.8 + 3.3 + 4.6 \approx 9.7$ clusters from the 3 observed.

No luminosity function is needed. We never compare counts at different luminosities. Each cluster is compared only with its own orbit, and the probability of catching it at a given phase depends on its orbit, its own (measured) M_V and C, not on how common clusters of its luminosity are. In a toy population where half of each orbit looks bright and half faint, the number of clusters in a luminosity group changes how many we see, not the near/far split of those we see. The corrected histogram of M_V is then an *output*: the luminosity function. The price is the assumption that replaces it: clusters are at random phases of their orbits today (phase mixing), whatever their luminosity.

The real analysis is the same procedure with all 165 clusters, 8 bins in V (7 unknown ratios, Eq. 2) and the likelihood maximised numerically. Two refinements (Sec. 5.8) handle faint clusters. A continuous $C(V)$ keeps falling beyond the faintest detection. And in the $1/V_{\rm max}$ form, a faint cluster's detection chance is averaged over the orbits of bright clusters, which are visible almost anywhere, rather than over its own orbit alone. This counts faint clusters on orbits where none has been seen, on the assumption that faint and bright clusters share one orbit distribution. On mocks matched to the observed sample, neither form is reliably better (Sec. 5.9), and we quote the simpler own-orbit form as the headline. Two further ingredients matter: extinction along each orbit (Secs. 5.6, 5.7) and, as a check, completeness that depends on direction (Sec. 5.10).

2 Data

We use the 165 confirmed GCs with full phase-space data (positions, distances, proper motions, line-of-sight velocities and their errors) from the PI's compilation of Vasiliev & Baumgardt (2021)

astrometry. Absolute magnitudes come from the Baumgardt combined table[1] as $M_V = 4.83 - 2.5\log_{10}(M_{\rm dyn}/\Upsilon_V)$, with $M_{\rm dyn}$ the dynamical mass and Υ_V the V-band mass-to-light ratio. Two confirmed clusters without 6D data are excluded. Progenitor labels (in-situ: main bulge and main disc, 61 GCs; accreted: Gaia-Enceladus, Sequoia, Helmi streams, high-energy group and Sagittarius, 64 GCs) come from the catalogue column `PROG_MA`; we have not verified their provenance (they appear to follow Massari et al. 2019).

Positions and velocities are transformed to the Galactocentric frame with the astropy v4.0 solar parameters ($R_0 = 8.122$ kpc, $z_\odot = 20.8$ pc, $\boldsymbol{v}_\odot = (12.9, 245.6, 7.78)$ km s^{-1}; Astropy Collaboration 2022). Orbits are integrated with AGAMA (Vasiliev 2019) in a static axisymmetric potential (Miyamoto–Nagai disc, $6.8\times10^{10}\,M_\odot$, $a = 3.0$, $b = 0.28$ kpc; two Dehnen spheroids, $5\times10^{9}\,M_\odot$ at 1.0 kpc and $1.71\times10^{9}\,M_\odot$ at 0.07 kpc; NFW halo, $5.4\times10^{11}\,M_\odot$, $r_s = 15.62$ kpc), forward for 5 Gyr from the present and sampled at 500 equal time steps. Energy is conserved to 2×10^{-6}. No observed cluster and no orbit sample lies beyond 150 kpc.

3 Method

3.1 Per-cluster phase likelihood

Let cluster i move on a known orbit, sampled at times t, with heliocentric distance $d_i(t)$. If the population is phase-mixed in a static potential, the present phase of an undetected cluster is uniform in t. Detection with probability C tilts that distribution, so the present phase of a *detected* cluster, conditional on its orbit, has density proportional to C at that phase. The log-likelihood of the observed present positions is

$$\ln\mathcal{L} = \sum_i \Big[\ln C(M_{V,i}, d_{i,\rm now}) - \ln\big\langle C(M_{V,i}, d_i(t))\big\rangle_t\Big]. \tag{2}$$

Two properties make Eq. (2) useful. First, no model of the intrinsic GC density enters. Second, conditioning on each detected orbit removes the dependence on which orbits happen to be in the catalogue. The overall scale of C cancels, so the data constrain C only up to one constant, and only over the range of d (or V) that the observed orbits span.

We take C piecewise constant in bins, write $\theta_k = \ln(C_k/C_0)$ with the reference bin $k = 0$ fixed, and maximise Eq. (2) with L-BFGS-B ($-8 \le \theta_k \le 4$). The likelihood is concave in θ. Intervals are 68% profile-likelihood intervals ($2\Delta\ln\mathcal{L} = 1$). The last bin is open-ended.

3.2 Two parameterisations of C

$C(M_V, d)$ per luminosity class. We split the sample into three equal-count M_V classes ($\approx$ 55 GCs each) and, within each, into equal-count distance bins of at least 15 GCs (three bins of 18–19 GCs). Bins are chosen once on the observed sample and then frozen. Each class yields $C(d)/C(\text{nearest bin})$; the classes are not linked to one another.

$C(V)$. To link the classes we assume completeness depends on apparent magnitude only, $C = C(V)$ with $V = M_V + 5\log_{10}(d/10\,\text{pc})$. This is a special case of $C(M_V, d)$: a faint cluster nearby and a bright cluster farther away with the same V share the same completeness. All 165 clusters then enter one fit, with $V_i(t)$ varying along each orbit. We use eight equal-count bins in observed V (20–21 GCs each) and fix the brightest bin ($V < 6.0$) as complete, $C_0 = 1$. This single anchor makes C absolute at all M_V. V excludes extinction; Sec. 5.6 tests one extinction model.

[1] Downloaded 2026-10-09 from https://people.smp.uq.edu.au/HolgerBaumgardt/globular/combined_table.txt.

3.3 Corrected counts

Each observed cluster receives the weight $w_i = 1/q_i$, with $q_i = \langle C(V_i(t))\rangle_t$ its orbit-averaged completeness. Corrected distributions are weighted histograms. This is the Horvitz–Thompson (1952) inverse-inclusion-probability estimator with the inclusion probability averaged over orbital phase. It corrects for incompleteness along the orbits of observed clusters; it cannot add clusters on orbit families that are never detectable. Uncertainties on corrected counts come from a cluster bootstrap (200 replicates): each replicate resamples clusters, refits C on the fixed bins, and sums the refitted weights of the resampled clusters.

3.4 Symmetry tests

If the model holds, the present position of each cluster is a draw from its orbit's C-weighted phase distribution. For three statistics — z, y (the Galactic-longitude side) and the azimuth from the Sun's direction $|\phi'|$ — we compute each cluster's randomised probability integral transform (PIT) against that distribution, pool the 165 values, and form a Kolmogorov–Smirnov statistic. The p-value comes from 500 null simulations that redraw present phases from the fitted C and refit C in every simulation, with a +1 correction.

3.5 Comparison estimator: orbit superposition with a fixed target

Following Khoperskov et al. (2025), we also fit non-negative orbit weights so that the time-averaged occupancy of the observed orbits on an $(R, |z|)$ grid reproduces a fixed target density (spherical broken power law with inner slope 2, outer slope 4, $r_b = 5$ kpc, $r_c = 0.3$ kpc; and variants), and take $C = N_{\rm obs}/N_{\rm corr}$ in each distance bin. Unlike Eq. (2), this estimator depends on the assumed target density.

4 Validation on mock catalogues

Mock populations are drawn from an isotropic quasi-spherical distribution function in the same potential, with a broken-power-law density (inner slope 2, outer slope 4, $r_b = 5$ kpc, $r_c = 0.3$ kpc), 250 GCs per realisation and a Gaussian luminosity function $M_V \sim \mathcal{N}(-7.4, 1.2)$. Each mock cluster sits at a uniformly random phase of its orbit and is detected with probability $C_{\rm true} = \{1+\exp[(M_V-(V_{\rm lim}-{\rm DM}))/0.5]\}^{-1}$, $V_{\rm lim} = 8$, so $C_{\rm true}$ is a function of V. Scenarios B–D break the assumptions: B multiplies C by a latitude term $1 - 0.7\,e^{-|b|/5^\circ}$; C adds a non-phase-mixed clump of 15 Sagittarius-like clusters; D makes C depend on M_V and d but not through V alone. Each estimator is compared with its own correct truth: for the phase likelihood, the value it converges to for an infinite sample; for the target estimator, the true detected-to-intrinsic ratio per bin.

Table 1: Mock recovery of relative completeness $C(d)/C(\text{nearest bin})$ per luminosity class (100 realisations per scenario). Ratios are recovered/true, median and 16–84% range; coverage is the fraction of nominal 68% intervals containing the truth.

Estimator	A: assumptions hold	B: latitude-dependent C	C: Sgr-like clump
Phase likelihood	0.99 (0.68–1.41); cov. 0.67	0.98 (0.71–1.52); 0.71	0.97 (0.68–1.34); 0.71
Target, plain ratio	1.29 (0.88–1.85); 0.48	1.30; 0.54	1.39; 0.45
Target, bias-reduced	1.23 (0.84–1.77); 0.50	1.22; 0.58	1.32; 0.47

Tables 1 and 2 and Figs. 1 and 2 summarise the tests. The phase likelihood recovers completeness without bias and with calibrated intervals in every scenario, including those that break its assumptions. The target estimator is biased high by about 25% even when given the correct target. The corrected totals are low: in scenario A, where the model is correct, they recover 80% of the true population and 44% of the faint clusters. The shortfall comes from faint clusters whose orbits never bring them within detectable distance; no reweighting of the observed orbits recovers them. The

Table 2: Mock validation of $C(V)$ and the corrected counts (100 realisations per scenario).

	A: $C(V)$ true	B: latitude	C: Sgr clump	D: not $C(V)$
$C(V)$ shape, recovered/true	0.92	1.08	1.01	1.08
68% coverage	0.69	0.68	0.68	0.67
Corrected total / true	0.80 (0.60–1.07)	0.56	0.71	0.87
Faint ($M_V > -6.5$): observed / true	0.15	0.12	0.15	0.64
Faint: corrected / true	0.44	0.27	0.42	0.73

bootstrap intervals on the total match the scatter across realisations (half-width $\approx 26\%$ of truth, scatter 24%), so their low coverage reflects this bias, not mis-sized intervals.

Two diagnostics have little power at this sample size. The symmetry tests reject at $p < 0.05$ in 6–13% of null mocks (the $|\phi'|$ test reaches $p \leq 0.01$ in 2%) and in 2–19% of mocks that break the assumptions. The AIC comparison of $C(V)$ against free per-class $C(d)$ favours the free model in only 31% of scenario-D mocks.

5 Results

5.1 Completeness per luminosity class

Table 3 and Fig. 3 give the per-class completeness. In the two brighter classes completeness stays near unity out to ≈ 10 kpc and falls to 0.4–0.6 beyond. The faint class drops to 0.44 already at 8–21 kpc. In-situ clusters lose completeness steeply with distance (0.18 beyond 8.8 kpc); accreted clusters stay near unity to 27 kpc and fall to 0.27 beyond (Fig. 4). The fixed-target estimator agrees with the phase likelihood within the intervals in most bins, but for the accreted clusters it gives values above 1 beyond 18 kpc, which the phase likelihood does not reproduce; these follow from the assumed target shape.

Table 3: Phase-likelihood completeness relative to the nearest distance bin, with 68% profile intervals. Each class holds 55 GCs in three bins of 18–19.

M_V class	nearest bin	second bin	third bin
-10.54 to -7.58	0–8.3 kpc: 1	8.3–11.1 kpc: 1.22 [0.88, 1.71]	$>$ 11.1 kpc: 0.63 [0.41, 0.95]
-7.58 to -6.50	0–7.4 kpc: 1	7.4–9.5 kpc: 0.75 [0.52, 1.07]	$>$ 9.5 kpc: 0.41 [0.28, 0.61]
-6.50 to -0.96	0–7.8 kpc: 1	7.8–21.5 kpc: 0.44 [0.31, 0.62]	$>$ 21.5 kpc: 0.56 [0.27, 1.22]

5.2 Symmetry tests

The pooled PIT tests (Fig. 5) give $p = 0.46$ for z, 0.042 for y and 0.006 for $|\phi'|$. Split by population, the in-situ clusters give $p = 0.040$, 0.29 and 0.006, and the accreted clusters 0.51, 0.49 and 0.75. Since null mocks reach $p \leq 0.01$ for $|\phi'|$ in only 2% of realisations, the in-situ clusters carry an azimuthal structure that completeness depending on (M_V, d) alone does not explain. The Galactic bar, which this analysis does not mask, is one candidate. The data do not identify the cause.

5.3 Completeness versus apparent magnitude

Table 4 and Fig. 6 give $C(V)$. Completeness falls steadily from unity at $V < 6$ to 0.14 [0.07, 0.30] at $V > 11$. The per-class completeness curves implied by $C(V)$ follow the free per-class fits (Fig. 6, right), and the AIC prefers $C(V)$ (-6.0 against -1.5), although the mocks show this comparison is weak.

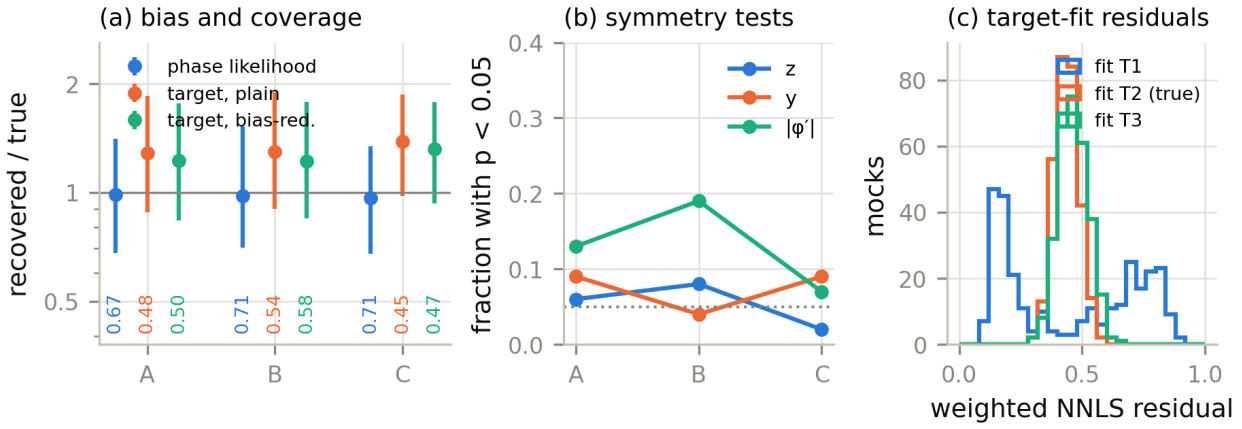


Figure 1: Mock validation of the per-class estimators. (a) Recovered/true relative completeness (median and 16–84% range) for the phase likelihood and the target estimator, plain and bias-reduced, in scenarios A–C; the vertical numbers give the 68% coverage. (b) Fraction of mocks in which each symmetry test gives $p < 0.05$; the dotted line marks 5%. (c) Target-fit residuals when the truth is T2; a residual cannot rank targets, since a T1 fit often beats the true T2.

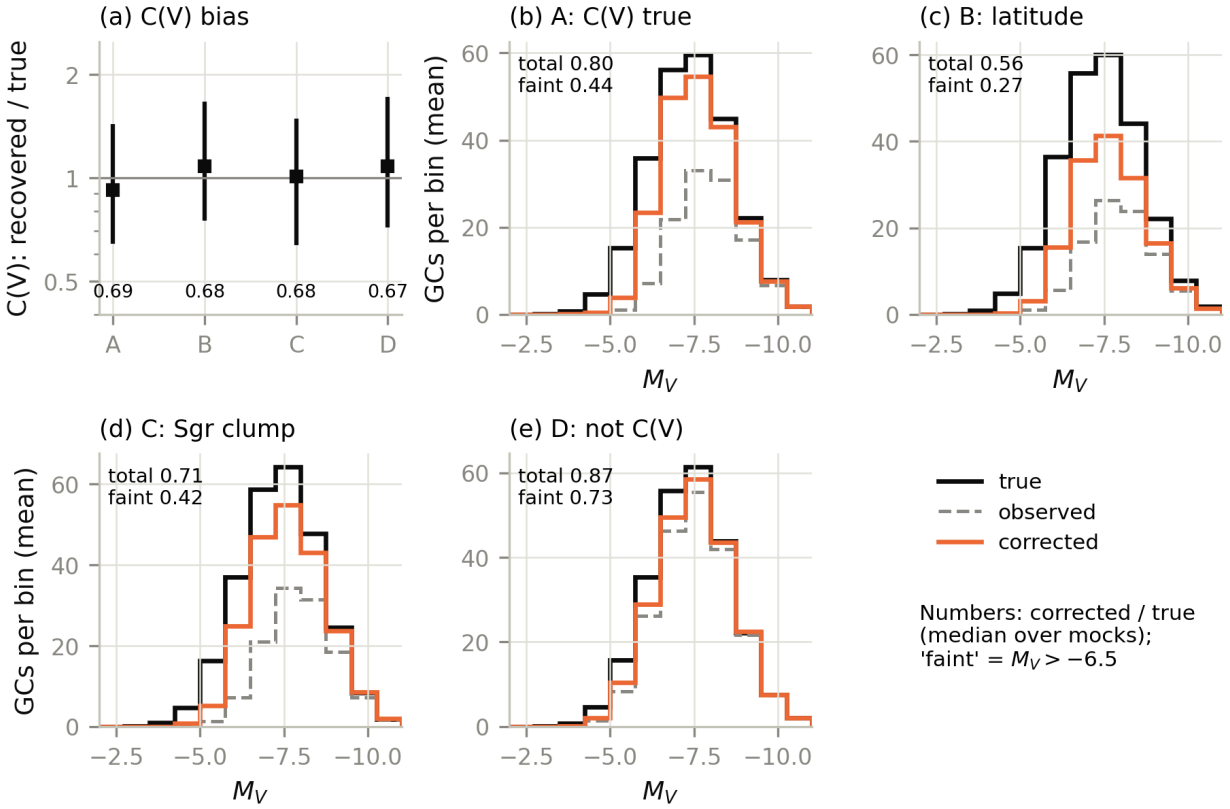


Figure 2: Mock validation of $C(V)$. (a) Recovered/true $C(V)$ per bin (median and 16–84% range) in scenarios A–D; numbers give the 68% coverage. (b)–(e) True (black), observed (dashed) and corrected (orange) M_V distributions, mean over realisations; numbers give the median corrected/true ratio for all GCs and for $M_V > -6.5$.

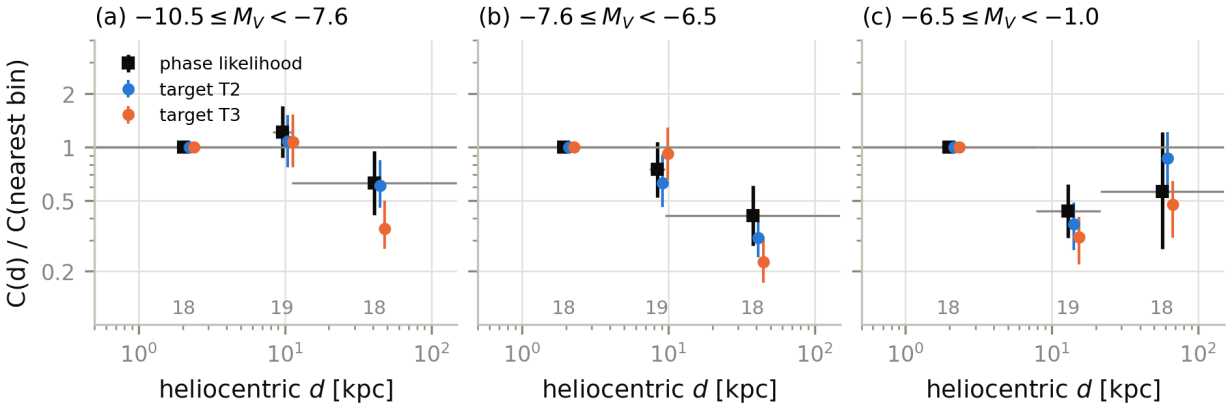


Figure 3: Completeness relative to the nearest distance bin for three M_V classes: phase likelihood (black squares, 68% profile intervals; grey bars show the bin extent) and the fixed-target estimator for targets T2 and T3 (coloured; bias-reduced, 68% bootstrap). Numbers give the observed clusters per bin.

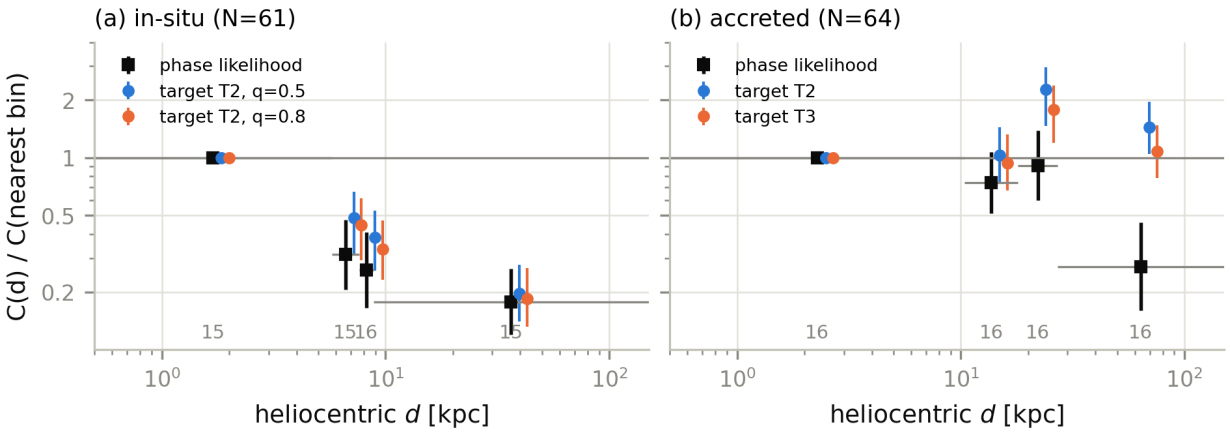


Figure 4: As Fig. 3, for in-situ (left) and accreted (right) clusters, all luminosities combined.

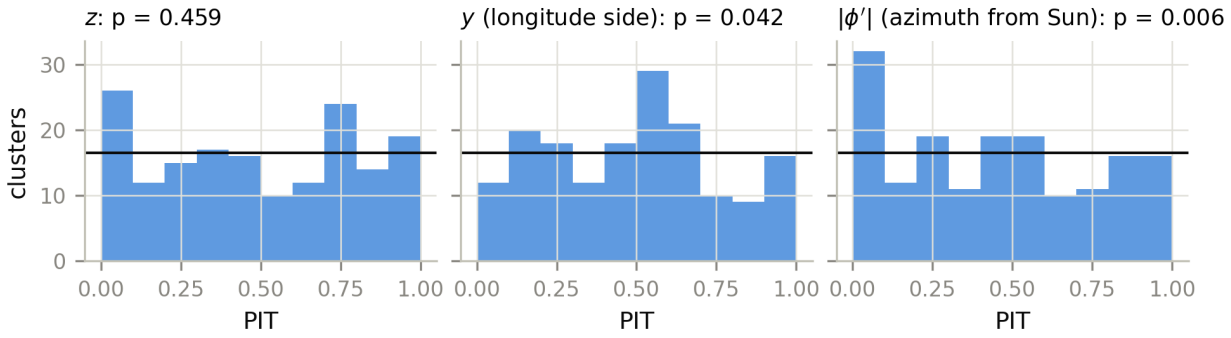


Figure 5: Pooled PIT values for z, y and $|\phi'|$ against each orbit's C-weighted phase distribution. Uniform histograms are expected if the model holds; the titles give the KS statistic and the null-calibrated p-value.

Table 4: $C(V)$ from the phase likelihood (brightest bin fixed at 1); 68% profile intervals. Each bin holds 20–21 GCs.

V	< 6.0	6.0–6.6	6.6–7.2	7.2–7.6	7.6–8.2	8.2–9.2	9.2–11.0	> 11.0
C	1	1.17	0.56	0.64	0.41	0.25	0.21	0.14
68%	–	$[0.82, 1.68]$	$[0.38, 0.82]$	$[0.43, 0.95]$	$[0.27, 0.62]$	$[0.16, 0.39]$	$[0.13, 0.35]$	$[0.07, 0.30]$

5.4 Corrected luminosity and metallicity distributions

Weighting by $1/q_i$ (weights from 0.94 to 6.9) turns the 165 observed GCs into 483 (68%: 335–765). The correction grows with faintness (Fig. 7): bright clusters ($M_V < -8$) rise from 40 to 60, intermediate ones ($-8 \leq M_V < -6.5$) from 70 to 156, and faint ones ($M_V \geq -6.5$) from 55 to 267 (168–473). The corrected metallicity distribution is about three times the observed one at all [Fe/H] and keeps its bimodal shape, with peaks near -1.5 and -0.6. Given Sec. 4, these corrected numbers are lower limits if the real selection resembles the mocks.

5.5 Why extinction matters, and what the mocks show

Whether a cluster is found depends on how bright it looks, and dust makes it look fainter: the relevant apparent magnitude is $V + A_V$, not V. Dust is concentrated in the Galactic disc, and most GCs live in the inner Galaxy, where many sightlines cross the disc and the bulge. The catalogue's present-day reddenings reach $E(B-V) = 6.8$, i.e. $A_V \approx 21$ mag, and half the clusters have $A_V > 1$ mag.

Extinction enters the phase likelihood in two places.

1. **The shape of C.** Along an orbit a cluster's brightness changes not only with distance but also with the sightline: passing behind the bulge it can dim by several magnitudes at almost the same distance. If A_V is left out, the fit can only attribute that dimming to distance or luminosity, which distorts $C(V)$.

2. **The corrected counts.** An orbit that spends time behind thick dust is detectable for a smaller fraction of its period. Ignoring the dust overestimates its orbit-averaged completeness q_i, so its weight $1/q_i$ is too small and the counts are under-corrected.

Because extinction depends on direction as well as distance, it also breaks the assumption that completeness depends on M_V and d alone, unless it is folded into V.

To test this we added a mock scenario (E) in which detection depends on $V + A_V$, with A_V computed along every orbit from an exponential dust disc (A_V of 1 mag kpc^{-1} in the plane at the solar circle, radial scale 3 kpc, vertical scale 0.1 kpc). We then fit each mock twice: with the true A_V added to V, and ignoring dust. Over 100 realisations:

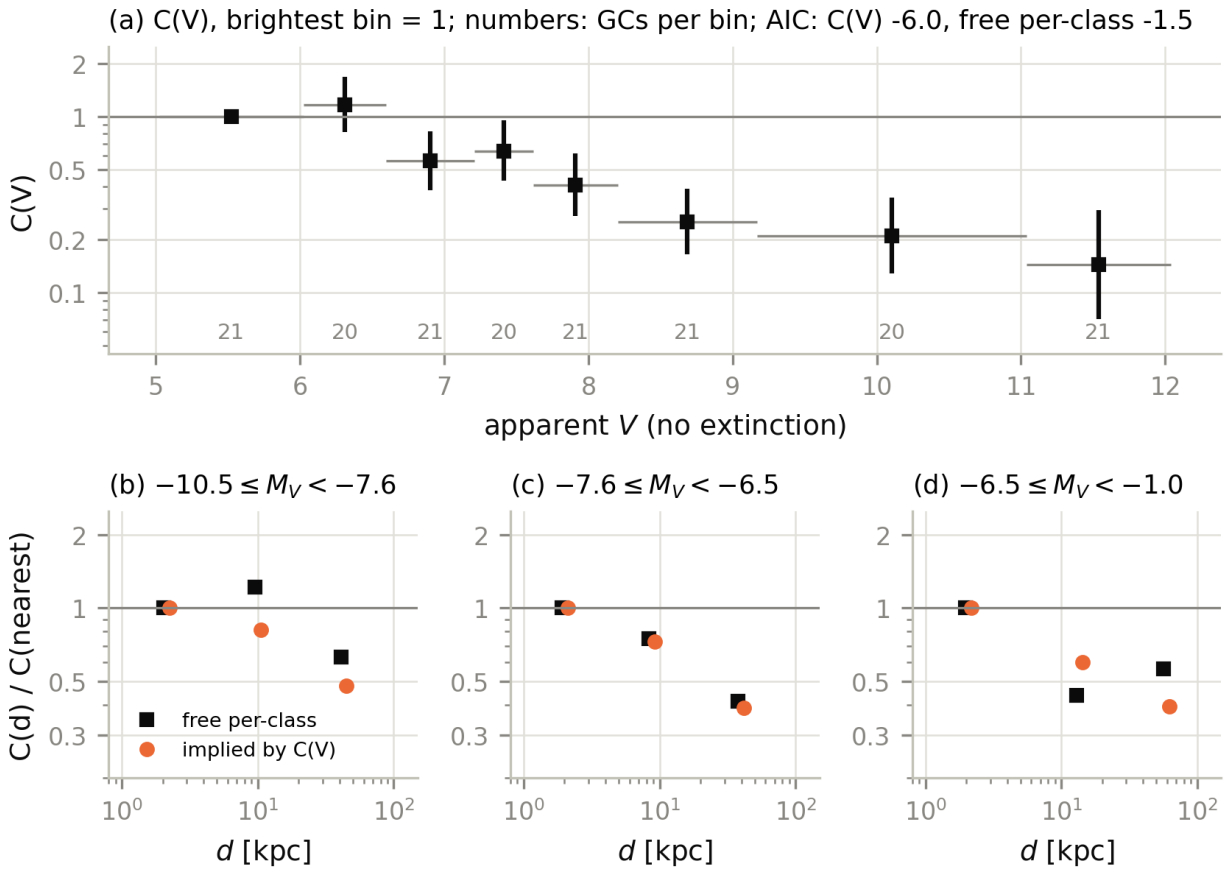


Figure 6: (a) $C(V)$ with 68% profile intervals; grey bars show the bin extent and numbers give clusters per bin. (b)–(d) Per-class completeness versus distance from free fits (black) and implied by $C(V)$ (orange).

- **The completeness curve is recovered when dust is included:** recovered/true $C(V+A_V)$ is 1.00, with 68% coverage 0.62.
- **Including dust reduces the bias in the corrected counts, but does not remove it:** the corrected total is 0.61 of the truth with the correct A_V and 0.51 ignoring dust; for faint clusters ($M_V > -6.5$) the figures are 0.18 and 0.16. The remaining shortfall is the unseen-orbit limitation of Sec. 4, which dust makes worse because it hides more orbits entirely: only 6% of the faint mock clusters are detected.
- **The likelihood can tell a correct dust model from no dust:** the AIC favours the true extinction model over ignoring dust in 98% of the realisations.

The last point matters for the real data. The likelihood comparison is a sensitive test, so the preference for no extinction over the present-day-reddening model in Sec. 5.6 is evidence against that particular model, rather than against the importance of dust. Separately, mock scenario B, whose latitude-dependent selection mimics dust that the model does not include, biases the corrected total low (0.56 of the truth).

5.6 Extinction: present-day reddening, 3D maps, and simple proxies

We tested three treatments of dust, each compared with the no-dust $C(V)$ fit by AIC (Table 5).

Present-day reddening held fixed. Adding $A_V = 3.1\,E(B-V)$, with each cluster's catalogue reddening at its present position held fixed along its whole orbit, raises the totals to 689 (457–1172)

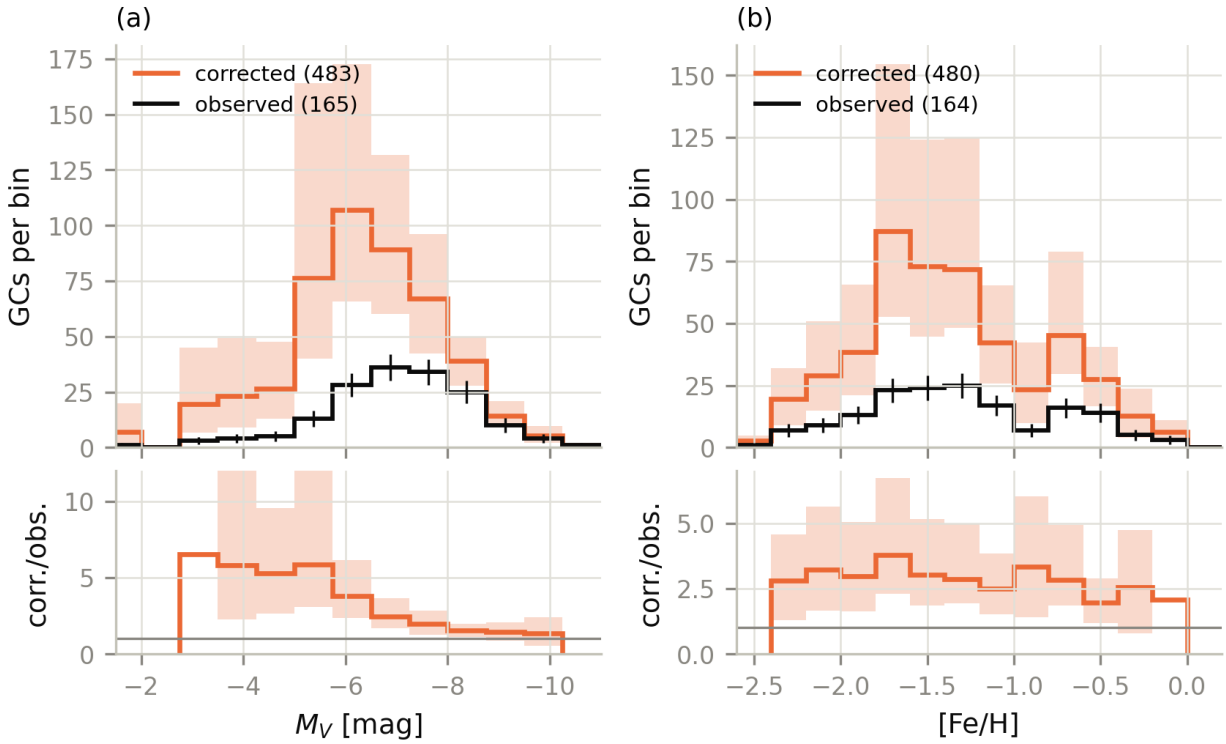


Figure 7: Observed (black) and completeness-corrected (orange, 68% bootstrap band) M_V and [Fe/H] distributions using $C(V)$. Lower panels: corrected/observed.

and 337 (203–651) faint. The data disfavour this model (AIC +1.3 against −6.0 without dust). Holding present reddening fixed while a cluster moves to other sightlines is unphysical; this test shows only that the total is sensitive, by about 43%, to an untested prescription.

3D dust maps along each orbit. We evaluate extinction at every one of the 165 × 500 orbit samples with the `dustmaps` package. The maps are combined by priority: DECaPS 3D (Zucker et al. 2025; southern plane $239° < l < 6°$, $|b| < 10°$, reliable to 10 kpc), then Bayestar19 (Green et al. 2019; where flagged reliable), then Marshall et al. (2006; inner plane, within each sightline's last distance slice), and SFD (Schlegel et al. 1998) otherwise. DECaPS supplies A_V for 24% of the orbit samples, Marshall for 13%, Bayestar19 for 8% and SFD for the remaining 55%, mostly at high latitude. Conversions: $A_V = 3.1\,E(B-V)$ for DECaPS and Bayestar19 ($E(B-V) = 0.883\times$ the Bayestar19 value), $A_V = 8.8\,A_{K_s}$ for Marshall, and $A_V = 2.742\,E(B-V)_{\mathrm{SFD}}$ (Schlafly & Finkbeiner 2011). Beyond a 3D map's reliable distance the true A_V is bracketed: the low bracket keeps the map's last value (no more dust), the high bracket uses the SFD total of the sightline (all its dust in front). At the clusters' present positions the maps give 5–8% less extinction than the catalogue reddening (median map/catalogue ratio 0.92 and 0.95 for the low and high brackets, 144 clusters with catalogue $A_V > 0.1$).

This model fits the data best (AIC −13.7 and −14.4 for the two brackets, against −6.0 without dust). Since the mocks show the likelihood picks the correct dust model in 98% of realisations (Sec. 5.5), the improvement is meaningful. The corrected total *falls* to 411–455, with tighter intervals, and the faint count to 197–218 (Fig. 8). Accounting for dust lets distant clusters behind it count as dimmed rather than distant, which changes the shape of C. With dust, the correction for metal-poor clusters ([Fe/H] < -1) drops from ×3.1 to ×2.4, while that for metal-rich clusters stays at ×2.6, so the two now receive similar corrections.

Simple obscuration proxies. Without 3D maps, obscuration can be fitted directly: we write $C = C_V(V)\,g(x)$, with x either $|b|$ or the 2D SFD reddening of the sightline at each orbit sample,

and g a step function in three equal-count bins anchored to 1 in the least obscured bin. The phase likelihood learns g from where along their orbits the clusters are caught. On mocks (Table 6) the proxies do no harm when there is no obscuration and reduce the shortfall when there is; the AIC prefers them over $C(V)$ in 12–15% of null mocks and in 69–71% of the dust mocks. On the real data, $g(|b|)$ fits better than no dust (AIC -10.4), with clusters in the lowest-$|b|$ third detected about half as often ($g = 0.49$), and gives a corrected total of 520; the 2D-SFD proxy is weaker (AIC -7.5; total 462).

Table 5: Real data: dust treatments compared. AIC is relative to the same data (lower is better); totals and faint counts ($M_V > -6.5$) with 68% bootstrap intervals.

Model	AIC	Corrected total	Faint
No extinction, $C(V)$	-6.0	483 (335–765)	267 (168–473)
Present $E(B-V)$ fixed along orbit	$+1.3$	689 (457–1172)	337 (203–651)
Proxy $C_V(V)\, g(E(B-V)_{\rm SFD})$	-7.5	462 (346–668)	251
Proxy $C_V(V)\, g(\lvert b\rvert)$	-10.4	520 (400–753)	269
3D maps along orbits, low bracket	-13.7	455 (354–634)	218 (158–327)
3D maps along orbits, high bracket	$\mathbf{-14.4}$	411 (319–566)	197 (140–291)

Table 6: Mocks: corrected total / true (faint in brackets), median over 100 realisations, for $C(V)$ alone and the two proxy models. In scenario E, the $C(V)$ column ignores dust; with the true A_V supplied, the total is 0.61 (faint 0.18).

Scenario	$C(V)$	$C_V(V)\, g(\lvert b\rvert)$	$C_V(V)\, g$(2D column)
A: no obscuration	0.80 (0.44)	0.83 (0.43)	0.82 (0.41)
B: latitude-dependent selection	0.56 (0.27)	0.64 (0.31)	0.65 (0.31)
E: dust disc	0.51 (0.16)	0.68 (0.19)	0.69 (0.19)

5.7 Why the no-dust fit overcorrects: the inner Galaxy's dust applied to the halo

Adding dust to the model *lowers* the corrected total (483 to 411–455 binned, 749 to 432–461 continuous), although dust makes clusters fainter. The reason is that the no-dust fit is not free of dust: it absorbs the dust losses of the inner Galaxy into $C(V)$ and then applies them in every direction.

Where the signal comes from. If a cluster is detected with equal probability all along its orbit and is phase-mixed, the fraction u_i of its orbit time spent closer to the Sun than it is now is uniform on $[0, 1]$. For the 43 GCs at $d > 15$ kpc the median u is 0.44 and a KS test against uniform gives $p = 0.79$: within their own orbits, the halo clusters show no sign of incompleteness. For the 122 GCs at $d < 15$ kpc the median is 0.32 ($p = 6 \times 10^{-9}$): they are caught preferentially at the near parts of their orbits. These are the inner clusters whose orbits cross the disc and bulge. With the 3D maps, the 68 clusters whose orbits have a median galactocentric radius below 4 kpc have an orbit-averaged A_V of 3.6 mag (median) and spend 39% of their time behind $A_V > 3$. At their present positions the median is only 1.7 mag: they are caught in the less obscured parts of their orbits.

How the no-dust fit uses it. Without A_V, the likelihood can attribute that deficit only to V, so $C(V)$ falls steeply over $V \approx 7$–11, the range the inner orbits span. That steep curve also applies to halo clusters at the same V, on dust-free sightlines. Halo clusters whose whole orbit lies at $V > 11$ (Pal 3, Pal 4, Crater, AM 4, Whiting 1 and others) all get the largest weight,

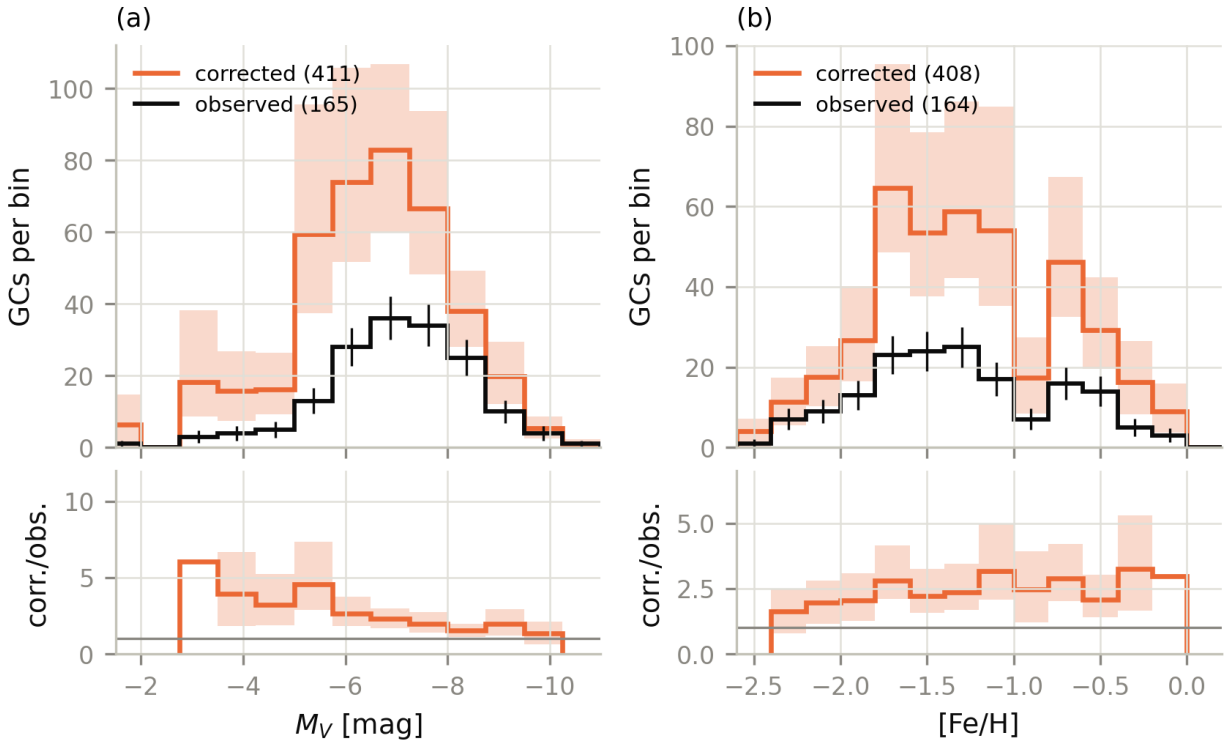


Figure 8: As Fig. 7, with A_V from 3D dust maps evaluated along every orbit (high bracket, the best-fitting binned model).

$1/C_{\rm last} = 1/0.144 = 6.9$. With the maps the curve becomes shallower, their q rises to 0.18–0.20 and their weights fall to 5.0–5.5, while the dust loss lands on the orbits that actually cross the dust.

Table 7 splits the change into two steps. Refitting C with the maps but switching the dust off along the orbits shows how much loss the no-dust curve had absorbed: all weights fall. Switching the dust back on restores the loss where it belongs: the inner weights double, the outer ones partly recover. Net, the inner clusters gain 19 and the outer ones lose 89. By present extinction: clusters with $A_{V,\rm now} < 1$ (84) go from a summed weight of 296 to 207, those with $A_{V,\rm now} > 1$ (81) from 187 to 204; the most obscured bulge clusters change most (Terzan 5, q from 1.00 to 0.21; Terzan 1, 0.96 to 0.16).

Table 7: Summed weights $\sum 1/q_i$ (binned C) for clusters whose orbits have median galactocentric radius below 4 kpc (inner) or above (outer). (a) no-dust fit $C_0(V)$; (b) C refitted with map extinction (high bracket) but evaluated without A_V along the orbits; (c) the map fit, $C_d(V + A_V)$.

	Inner (68)	Outer (97)	Total
(a) $C_0(V)$, no dust	152	330	483
(b) $C_d(V)$, dust off along orbits	86	200	287
(c) $C_d(V + A_V)$	171	241	411

Why mock E behaves differently. In mock scenario E, ignoring dust lowers the corrected total (0.51 against 0.61 of truth with the true A_V). The sign of the bias depends on the balance between the two terms: the extra loss on dusty orbits, which ignoring dust misses, and the steepened $C(V)$ applied to dust-free orbits at the same V. In the mock population the first dominates; in the real catalogue, where the largest weights belong to distant halo clusters, the second does. The likelihood settles which model is right in either case: it prefers the map model by ΔAIC = 8 (binned) and 16–17 (continuous), and in mocks it picks the true dust model in 98% of realisations.

5.8 Faint clusters on never-detectable orbits: continuous $C(V)$ and $1/V_{\rm max}$

Two sources of the faint-end shortfall. Mock tests isolate two causes of the low corrected counts in Table 2. With the true C, the own-orbit weights $1/q_i$ recover 87% of the total and 63% of the faint clusters (scenario A, median of 8 realisations): the rest are clusters whose q_i is nearly zero. With the fitted binned C the faint fraction falls to 41%. The last V bin is open, so the binned model holds C constant at all fainter V, where the true C falls to zero. This inflates q_i for faint clusters, whose orbits spend most of their time beyond the faintest detection. Even the best binned C (its pseudo-true value) recovers only 36% of the faint clusters.

Continuous $C(V)$. We therefore also fit $\ln C$ as a piecewise-linear function of V, with nodes at the median observed V of the same eight equal-count bins. It is anchored to $C = 1$ at the brightest node and held flat at brighter V, and beyond the faintest node it continues with the last slope, constrained to be ≤ 0. The model has the same seven parameters as the binned one and is fitted with the same likelihood (Eq. 2).

$1/V_{\rm max}$ over the orbit population. We assume the orbit distribution of GCs does not depend on luminosity. Bright clusters ($M_V < -8$) are detectable along essentially all orbits, so their $1/q$-weighted orbits sample the orbit population. A cluster of absolute magnitude M placed at a random phase on that population is detected with probability

$$f(M) = \frac{\sum_{j:\, M_V<-8} q_j^{-1} \, \langle C(M+\mu_j(t))\rangle_t}{\sum_{j:\, M_V<-8} q_j^{-1}}, \qquad \mu_j(t) = 5\log_{10}\frac{d_j(t)}{10\,\mathrm{pc}} + A_{V,j}(t), \tag{3}$$

and each observed cluster gets weight $1/f(M_{V,i})$ in place of $1/q_i$. This is the $1/V_{\rm max}$ estimator (Schmidt 1968) with the survey volume replaced by the orbit population. It counts faint clusters on orbits that no observed faint cluster occupies. For the split by metallicity we apply the assumption within each class ([Fe/H] < -1, ≥ -1), using the bright clusters of the same class (31 and 9) as its orbit population.

Mock validation. Table 8 adds two scenarios whose true $C(V)$ is not log-linear in the tail: F, a shallow all-sky survey plus a deep partial-area survey, $C = 0.8\,\ell(8, 0.3) + 0.2\,\ell(13, 0.3)$ with $\ell(V_0, w) = [1 + e^{(V-V_0)/w}]^{-1}$; and G, a power-law tail, $C = (1 + V - 7)^{-2}$ for $V > 7$. The continuous model with $1/V_{\rm max}$ recovers 0.89–1.13 of the true total and 0.80–1.03 of the faint clusters in every scenario that satisfies its assumptions (A, C, D, F, G). In scenario E, with the true A_V supplied, it recovers 0.95 of the total and 0.68 of the faint clusters. It fails, as expected, under the unmodelled latitude selection of B (0.73 and 0.65). The price is variance: across realisations the total ranges over 0.67–1.55 of truth (16–84%, scenario A), and the 68% bootstrap intervals cover the truth in 53–70% of mocks, so they are somewhat too narrow. The continuous model is preferred by AIC in 55–100% of mocks.

Table 8: Mocks: corrected total / true (faint, $M_V > -6.5$, in brackets), median of 100 realisations. Binned: current $C(V)$ with $1/q$. Continuous: $\ln C$ piecewise linear with an extrapolated tail, with own-orbit $1/q$ or with $1/V_{\rm max}$. In E, the true A_V is supplied.

Scenario	Binned, $1/q$	Continuous, $1/q$	Continuous, $1/V_{\rm max}$
A: $C(V)$ logistic	0.80 (0.44)	1.00 (0.83)	1.01 (0.95)
B: latitude term	0.56 (0.27)	0.74 (0.52)	0.73 (0.65)
C: Sgr-like clump	0.71 (0.42)	0.96 (0.76)	1.06 (1.03)
D: C not a function of V	0.87 (0.73)	0.88 (0.72)	0.89 (0.81)
E: dust disc	0.61 (0.18)	0.85 (0.50)	0.95 (0.68)
F: two surveys	0.94 (0.79)	1.12 (1.05)	1.13 (0.98)
G: power-law tail	0.75 (0.39)	1.04 (0.92)	0.91 (0.80)

Real data. The continuous model fits better than the binned one: AIC −22.6 against −14.4 with the map extinction (high bracket), −23.7 against −13.7 (low bracket), and −6.9 against −6.0 without extinction. Table 9 gives the corrected counts. With the maps, $1/V_{\max}$ gives a total of 393 (274–573) to 418 (293–625) and 177–189 faint clusters, close to the binned estimates of Table 5 (411–455; 197–218). By metallicity, the per-class $1/V_{\max}$ gives 245–259 metal-poor and 160–175 metal-rich clusters, against 289–320 and 120–133 with the binned model. The bright metal-rich clusters lie on inner, dusty orbits, so a faint metal-rich cluster on such an orbit is rarely detectable (Fig. 9).

Table 9: Real data, continuous $C(V)$: corrected counts with 68% bootstrap intervals. Faint: $M_V \geq -6.5$. $1/V_{\max}$ by [Fe/H] uses the bright clusters of the same class as orbit population.

	No extinction	Maps, low	Maps, high
AIC (binned)	−6.9 (−6.0)	−23.7 (−13.7)	−22.6 (−14.4)
Total, own-orbit $1/q$	749 (467–1185)	461 (338–685)	432 (321–628)
Total, $1/V_{\max}$	563 (348–841)	418 (293–625)	393 (274–573)
Faint, own-orbit $1/q$	527 (270–910)	224 (149–345)	208 (137–314)
Faint, $1/V_{\max}$	322 (175–508)	189 (119–292)	177 (112–274)
[Fe/H] < -1, $1/V_{\max}$ per class	425 (265–645)	259 (176–346)	245 (165–327)
[Fe/H] ≥ -1, $1/V_{\max}$ per class	119 (75–169)	175 (102–277)	160 (94–251)

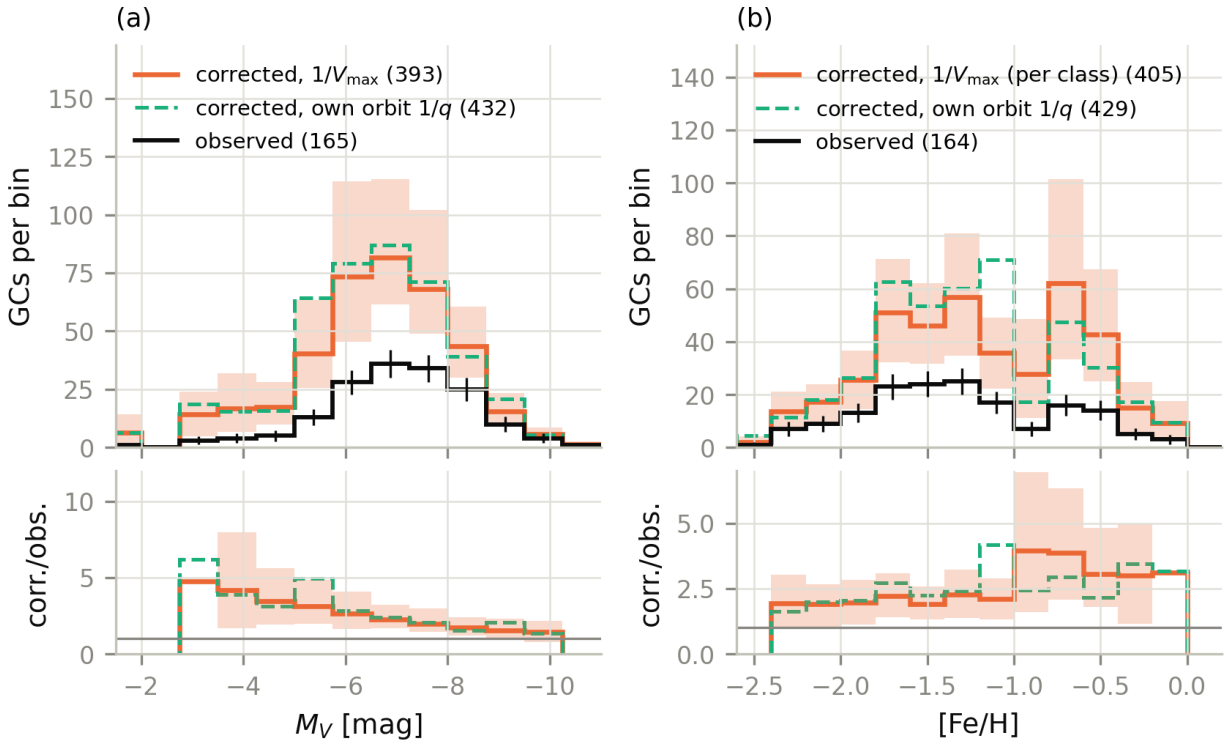


Figure 9: Corrected distributions with the continuous $C(V)$ and map extinction (high bracket): $1/V_{\max}$ (solid; band: 68% bootstrap) and own-orbit $1/q$ (dashed). (a) M_V, with the bright clusters of all metallicities as orbit population. (b) [Fe/H], with $1/V_{\max}$ computed per class.

5.9 Which faint-end weighting? Realistic mocks and the shared-orbit assumption

The question. $1/V_{\max}$ assumes faint and bright GCs share one orbit distribution. If they do, the detected faint clusters sit preferentially on the more detectable orbits, and the ratio $R =$

faint($1/V_{\rm max}$)/faint($1/q$) exceeds 1: its median is 1.0–1.4 in the mocks of Table 8. On the real data $R = 0.85$ (maps) and 0.61 (no extinction): the observed faint clusters lie on orbits *less* detectable than those of bright clusters. If faint GCs are preferentially on extended orbits, $1/V_{\rm max}$ undercounts them, and the mock-based preference for it does not carry over. We therefore audited R, built mocks matched to the observed sample, and compared four weightings on them, with acceptance tolerances fixed in advance (in `configs/cv.yaml`): $|\text{median } \ln(N_{\rm corr}/N_{\rm true})| \leq 0.10$ for totals and ≤ 0.15 for faint clusters, and 68% coverage in $[0.59, 0.77]$. The four weightings are

- own-orbit $1/q$;
- $1/V_{\rm max}$ with the bright clusters ($M_V < -8$) as orbit population;
- a luminosity-matched $1/V_{\rm max}$, in which the population is all clusters weighted by $1/q_j$ and by a Gaussian of width 1 mag in M_V;
- $1/V_{\rm max}$ with the intermediate class ($-8 \leq M_V < -6.5$) as orbit population for faint targets.

Audit of R on the real data. With the maps (high bracket), $R = 0.85$ with a paired bootstrap interval of 0.74–0.96; it is below 1 in 92–94% of resamples. Resampling the spline node placement as well gives 0.73–0.96. Three variants of the faint-end tail (an extra node at the 97.5th percentile of observed V, a flat tail, a tail twice as steep) and the per-[Fe/H] orbit populations give $R = 0.77$–0.89; the low map bracket gives 0.79–0.89. The bright orbit population is not dominated by a few clusters: its effective size is 31 of 40 and its largest normalised weight 8%. So $R < 1$ is not an artefact of the tail form or of a few weights.

Apocentres. The raw median apocentres of faint and bright GCs are 10.3 and 9.0 kpc ($\Delta \ln r_{\rm apo} = 0.13$, bootstrap -0.39 to 0.59). Weighted by $1/q$ they are 14.1 and 5.8 kpc ($\Delta \ln r_{\rm apo} = 0.89$, -0.05 to 1.34; positive in 82% of resamples). The weighting itself favours extended orbits for faint clusters, so only mocks can say what these numbers imply. Neither statistic sees orbit families with no detected member.

Realistic mocks. Mocks with the logistic truth of Sec. 4, at the observed sample size, detect only 2–11 faint clusters against 55 real ones, because their logistic $C(V)$ falls far more steeply than the fitted one. The new mocks use as true completeness the continuous $C(V)$ fitted to the real data (maps, high bracket), a Gaussian luminosity function with mean -6.8 and width 1.5 mag, and 252 clusters, which gives 167 detected, 54 of them faint, in the null case. Luminosity is drawn first; each class (bright, intermediate, faint) is then sampled from its own isotropic DF with break radius scaled by 1, $\sqrt{s}$ and s. $s = 1$ is the null (orbits independent of M_V); $s = 2$ and $s = 4$ make faint GCs progressively more extended. We add the dust disc of scenario E, and two misspecification controls: the logistic $C(V)$ and the power-law tail. Each scenario has 100 calibration and 100 validation realisations; the mock design was fixed after a smoke test that showed detection counts only.

Results (Table 10).

- **The intervals are calibrated once the mocks resemble the data.** The 68% bootstrap intervals (node placement resampled) cover the truth in 63–79% of validation realisations for every weighting and scenario. The factor that would make them cover exactly 68% on the calibration set is 0.92–1.03, so no inflation is needed. The 53–70% coverage of Sec. 5.8 came from the small, steeply selected samples of the earlier mocks.
- **No weighting is reliably better.** All four are within ± 0.17 in median ln in the realistic scenarios, and the scatter of a single sample is large (rms ln 0.36–0.61 for totals, 0.44–0.74 for faint clusters). The standard error of each median bias is 0.04–0.07, comparable to the differences between weightings, and on the same realisations the intermediate-class version changes $|\ln|$ errors by at most 0.04 relative to the others, with 95% intervals that include zero

Table 10: Validation mocks: median $\ln(N_{\rm corr}/N_{\rm true})$ for the total and for faint clusters ($M_V \geq -6.5$), 100 realisations per scenario. Columns: own-orbit $1/q$; $1/V_{\rm max}$ with the bright orbit population; luminosity-matched (kernel) $1/V_{\rm max}$; $1/V_{\rm max}$ with the intermediate class as population for faint targets. Last two rows: misspecification controls.

Scenario	$1/q$: total, faint	$1/V_{\rm max}$: total, faint	kernel: faint	intermediate: faint
Null ($s = 1$), real C	-0.06, -0.12	-0.14, -0.16	-0.12	-0.13
Null + dust disc	$+0.05$, $+0.10$	$+0.02$, $+0.01$	$+0.09$	$+0.04$
$s = 2$	$+0.06$, $+0.07$	-0.08, -0.17	$+0.02$	-0.04
$s = 4$	$+0.13$, $+0.15$	-0.03, -0.14	$+0.09$	-0.02
$s = 4$ + dust disc	$+0.12$, $+0.13$	$+0.02$, -0.01	$+0.10$	$+0.05$
Null, logistic C	-0.13, -0.35	-0.08, -0.16	-0.31	-0.19
Null, power-law tail	$+0.12$, $+0.15$	$+0.00$, -0.09	$+0.14$	-0.01

in nearly every scenario. Its apparent advantage in the $s > 1$ scenarios is also favoured by the mock design, whose orbit trend changes at exactly its class boundaries. The bright-population $1/V_{\rm max}$ is biased low even in the null (-0.14, -0.16).

- **The misspecification controls fail for all weightings** at the faint end (-0.16 to -0.35 for the logistic truth), as expected when $C(V)$ is poorly represented.

Do faint GCs have more extended orbits? Table 11 compares the real values of R and of the apocentre differences with their mock distributions. The real data fit a mild trend ($s = 2$) best. A strong trend ($s = 4$) is disfavoured: only 2–4% of those mocks have a raw apocentre difference as small as observed. No trend is not excluded: each statistic alone has a probability of 0.1–0.2 under the null, and R and the weighted apocentre difference jointly 0.02–0.09.

Table 11: Fraction of validation and calibration mocks (200 per scenario) at least as extreme as the real data: $R \leq 0.85$; $1/q$-weighted $\Delta \ln r_{\rm apo} \geq 0.89$; raw $\Delta \ln r_{\rm apo} \geq 0.13$.

Scenario	$R \leq 0.85$	weighted $\Delta \ln r_{\rm apo} \geq 0.89$	raw $\Delta \ln r_{\rm apo} \geq 0.13$
Null ($s = 1$)	0.16	0.10	0.18
Null + dust disc	0.16	0.02	0.21
$s = 2$	0.40	0.30	0.79
$s = 4$	0.64	0.74	0.98
$s = 4$ + dust disc	0.44	0.68	0.96

Choice. We take own-orbit $1/q$ as the headline weighting: it needs no choice of orbit population, passes in the null and mild-trend mocks, and is the closest to unbiased in the case the data favour. The other three give the sensitivity range. With the continuous $C(V)$ and the maps (high bracket) the corrected total is 432 (321–628) with $1/q$ and 393–432 across the four weightings; faint clusters number 208 (137–314) and 177–208. The low map bracket raises all values by 6–8%.

5.10 Direction-dependent completeness

Model. Sec. 5.7 shows that a single $C(V)$ can carry the selection of one part of the sky into another. We therefore let completeness depend on direction: each orbit sample is assigned to the inner window $|l| < 30°$, $|b| < 15°$ (fixed before fitting) or to the rest of the sky, and $\ln C$ is evaluated with that region's curve. The rest of the sky carries the anchor. We fit two forms: a shared continuous curve plus a constant inner contrast (one extra parameter), and separate continuous curves per region, with nodes at the median present V of the clusters now in each region. The inner level is constrained by orbits that cross between regions: 86 clusters spend 10–90% of their orbit in

each, while 39 stay in the window throughout. Both weightings carry over unchanged, with every orbit sample using its region's C.

Mocks. We added three scenarios to those of Sec. 4, 100 realisations each: H, a shallower limit inside the window ($V_{\rm lim} = 6$ against 8 outside); I, a constant factor 0.5 inside the window; and J, a direction effect in a larger window ($|l| < 60°$, $|b| < 25°$) than the one fitted. Table 12 gives the bias and scatter of the total.

- **The contrast model does no harm when there is no direction effect** (A, F, B, E, C: paired ratio to the single curve 1.00–1.01), and the AIC prefers it falsely in 14–26% of those mocks, close to the 16% expected for one extra parameter.
- **It detects a direction effect inside the window** (AIC preference 98% in H, 81% in I) and removes most of the bias (H: −0.10 to +0.01; I: −0.16 to −0.05).
- **It does not fix a direction effect outside the fitted window** (J: +0.17).
- **The separate curves have 30% less scatter**, but a control fit shows this comes from their fewer nodes per region: a single curve with as few nodes has the same scatter but is biased by +0.2 to +0.4 in H, J and E, where the separate curves are not.

Table 12: Mocks with direction-dependent completeness: mean and rms of $\ln(N_{\rm corr}/N_{\rm true})$ for the total ($1/q$), 100 realisations per scenario, and the fraction of mocks in which the AIC prefers the contrast model over the single curve.

Scenario	Single $C(V)$	Shared + contrast	Separate curves	AIC: contrast
A: no direction effect	−0.04 / 0.51	−0.03 / 0.51	+0.06 / 0.35	15%
F: two surveys	+0.10 / 0.51	+0.10 / 0.51	+0.15 / 0.36	15%
H: shallower limit in window	−0.10 / 0.51	+0.01 / 0.45	+0.03 / 0.39	98%
I: factor 0.5 in window	−0.16 / 0.48	−0.05 / 0.44	+0.13 / 0.37	81%
J: effect outside fitted window	+0.13 / 0.50	+0.17 / 0.51	+0.13 / 0.44	44%
B: latitude band	−0.11 / 0.50	−0.09 / 0.49	−0.04 / 0.34	26%
E: dust disc	−0.18 / 0.45	−0.18 / 0.44	+0.04 / 0.35	14%

Real data (Table 13, Fig. 10). We treat the regional models as a test of how sensitive the result is to direction-dependent selection, not as a substitute for the extinction maps.

- **Without extinction the data prefer direction dependence**: the separate curves are preferred by $\Delta{\rm AIC} = -4.2$, with the inner curve falling well below the outer one at faint V ($\ln C_{\rm in} - \ln C_{\rm out} = -1.65$ at $V = 12$). As expected from Sec. 5.7, the weights of clusters now in the window rise (summed $1/q$: 186 to 299), and the total rises from 749 to 802.
- **But the two-region split leaves most of the no-dust halo excess in place.** The summed $1/q$ of the 43 GCs at $d > 15$ kpc falls only from 453 to 376, against 135 with the maps, because the outer curve stays steep ($C_{\rm out} = 0.064$ at $V = 14$ without dust, 0.23 with the maps). Although 84% of the orbit samples with $A_V > 1$ lie inside the window, that steepness is not set by the halo clusters or by dusty sightlines outside the window. Decomposing $\partial \ln L/\partial\lambda$, where λ scales the outer curve's $\ln C$, into per-cluster terms shows that the pull towards a steeper outer curve comes from the 41 clusters now inside the window whose orbits leave it (+10.3, balanced by −9.5 from the 77 now outside whose orbits enter it). The halo clusters (−2.1) and the clusters with dusty off-window orbit segments (−2.3) pull the other way. The two regional curves are coupled through these crossing orbits: the extra loss of inner clusters that the model cannot attribute to dust is partly assigned to the outer curve, which then also applies to the halo. A third region (low latitude outside the window, $|b| < 15°$) removes more

of the excess (halo 257, total 575, $\Delta\mathrm{AIC} = -4.2$ against two regions) but not all of it. No single cluster dominates: in leave-one-out refits the ten most influential clusters carry 26% of the total influence on the halo weight.

- **With the 3D maps the data do not prefer extra direction dependence**: $\Delta\mathrm{AIC} = +1.0$ (contrast), 0.0 (two separate curves) and -1.1 (three regions). The remaining inner/outer contrast of the two-region fits is $\ln C_{\mathrm{in}} - \ln C_{\mathrm{out}} = -0.27$ (-0.53 to -0.02), a factor of about 0.76 at all V.
- **The totals move by -12 to $+12$%** (432 to 380–485 with $1/q$, maps high): 453 and 485 with two regions, 380 with three (halo weight 82). The intervals are no wider for the two-region models, with p_{84}/p_{16} 1.96–1.99 for all three. The three-region fit has only 19 clusters now in its low-latitude region and two nodes there, so we do not adopt it.
- A check of the model, the C-weighted version of the u test above, passes inside the window ($p \approx 0.6$) under every model, and fails mildly outside it ($p = 0.01$–0.025) under every model, with or without the maps. Neither a regional curve nor the maps account for that residual, so some mismatch between model and data remains possible.

Table 13: Real data: corrected totals with direction-dependent C; ΔAIC relative to the single continuous $C(V)$ with the same extinction; 68% bootstrap intervals.

	No extinction			Maps, high bracket		
Model	ΔAIC	$1/q$	$1/V_{\mathrm{max}}$	ΔAIC	$1/q$	$1/V_{\mathrm{max}}$
Single $C(V)$	0	749 (467–1185)	563 (348–841)	0	432 (321–628)	393 (274–573)
Shared + contrast	-0.9	732 (485–1261)	567 (384–875)	$+1.0$	453 (333–657)	408 (298–559)
Separate curves	-4.2	802 (592–1334)	671 (491–938)	$+0.0$	485 (361–719)	437 (316–598)

6 Summary of results and error budget

With continuous $C(V)$, 3D-map extinction (high bracket) and own-orbit weights, the 165 GCs with 6D data correct to 432 (68%: 321–628), about 2.6 times the observed number, of which 208 (137–314) are fainter than $M_V = -6.5$, against 55 observed. Table 14 lists how each choice moves the total. The refinements of the weighting, the faint tail and the direction dependence move it by up to 12%; the bootstrap interval spans -26 to $+45$%. The two changes that matter are the extinction along the orbits, a factor of 1.7, and the continuous $C(V)$, which in mocks fixed a factor-of-two undercount of faint clusters. The systematics in the last three rows are not yet quantified.

Table 14: Error budget for the corrected total (continuous $C(V)$, maps high bracket, $1/q$: 432).

Source	Range of the total	Relative
Sampling (cluster bootstrap, 68%)	321–628	-26%, $+45$%
Weighting ($1/q$, three $1/V_{\mathrm{max}}$ forms)	393–432	-9%
Faint-end tail shape (three variants)	424–457	-2%, $+6$%
Direction-dependent C (two or three regions)	380–485	-12%, $+12$%
Map bracket (high, low)	432–461	$+7$%
Extinction treatment (none vs maps)	749 vs 432–461	factor 1.7
Anchor (brightest V complete)	all weights scale as $1/C_{\mathrm{anchor}}$	not quantified
Phase mixing, static potential, orbit errors	—	not quantified
Clusters on never-detectable orbits	—	not identifiable

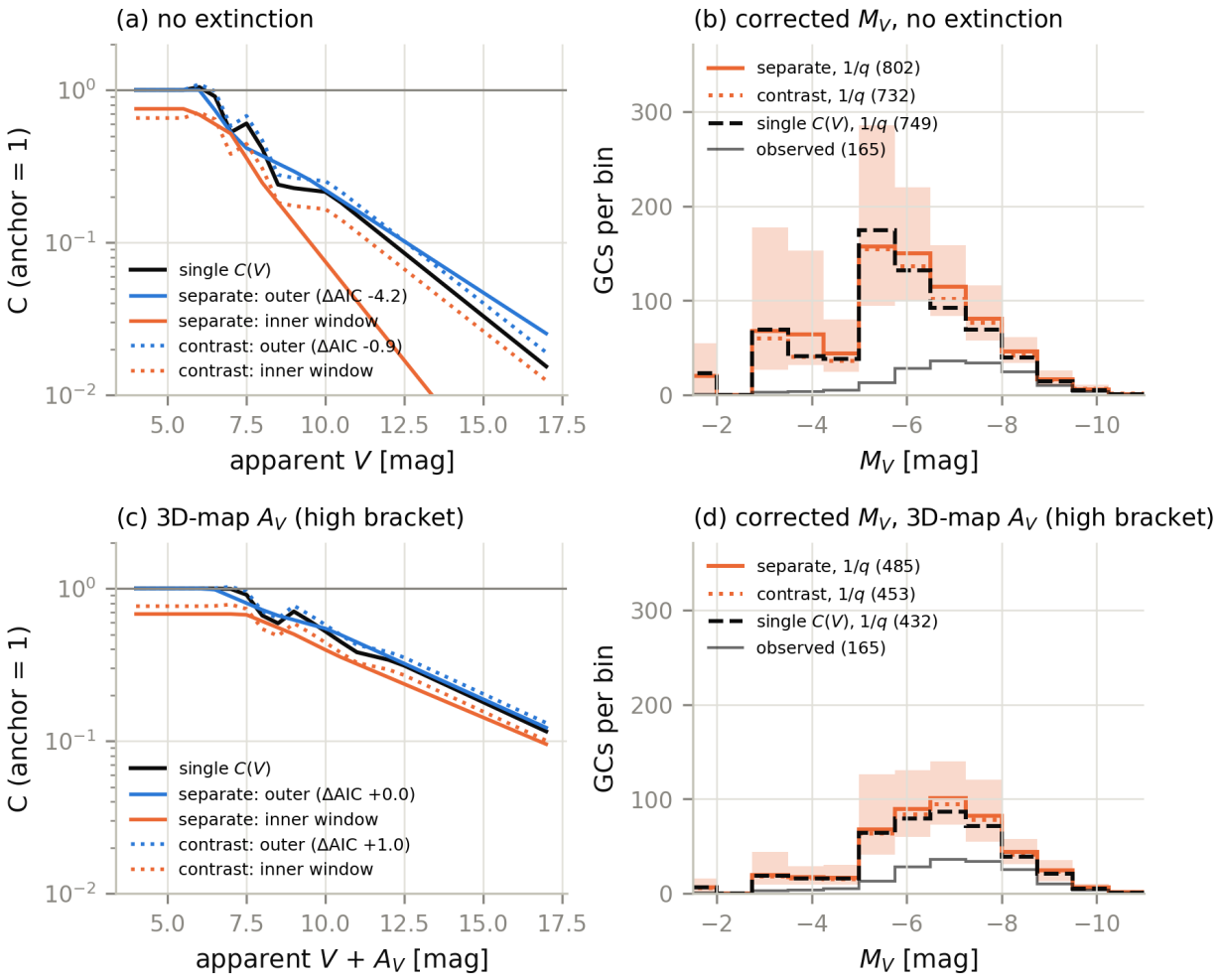


Figure 10: Direction-dependent completeness without extinction (top) and with map extinction (bottom, high bracket). (a, c) C for the single curve (black), and for the outer sky (blue) and the inner window $|l| < 30°$, $|b| < 15°$ (orange), with separate curves (solid) or a shared curve plus a constant contrast (dotted). (b, d) Corrected M_V distributions with $1/q$; band: 68% bootstrap for the separate curves.

6.1 The final answer

$$N_{\rm GC} \approx 430^{+215}_{-125} \qquad (68\%,\ \text{sampling and quantified systematics}), \tag{4}$$

about 2.6 times the 165 GCs with 6D data. Of these, about 210 (68%: 125–325, sampling and quantified systematics; 140–315 from sampling alone) are fainter than $M_V = -6.5$, against 55 observed (Fig. 11).

What the central value is. The total from the continuous $C(V)$ (Sec. 5.8), with extinction from 3D maps along every orbit (high bracket, Sec. 5.6) and each cluster weighted by the inverse of its own orbit-averaged completeness, $1/q_i$: 432. This is the model the likelihood prefers (Table 9); ignoring extinction (749) is rejected by $\Delta\text{AIC} \approx 16$ and biased high for the reason given in Sec. 5.7.

How the error bar is built. We work in $\ln N$ and take each component's largest excursion from Table 14:

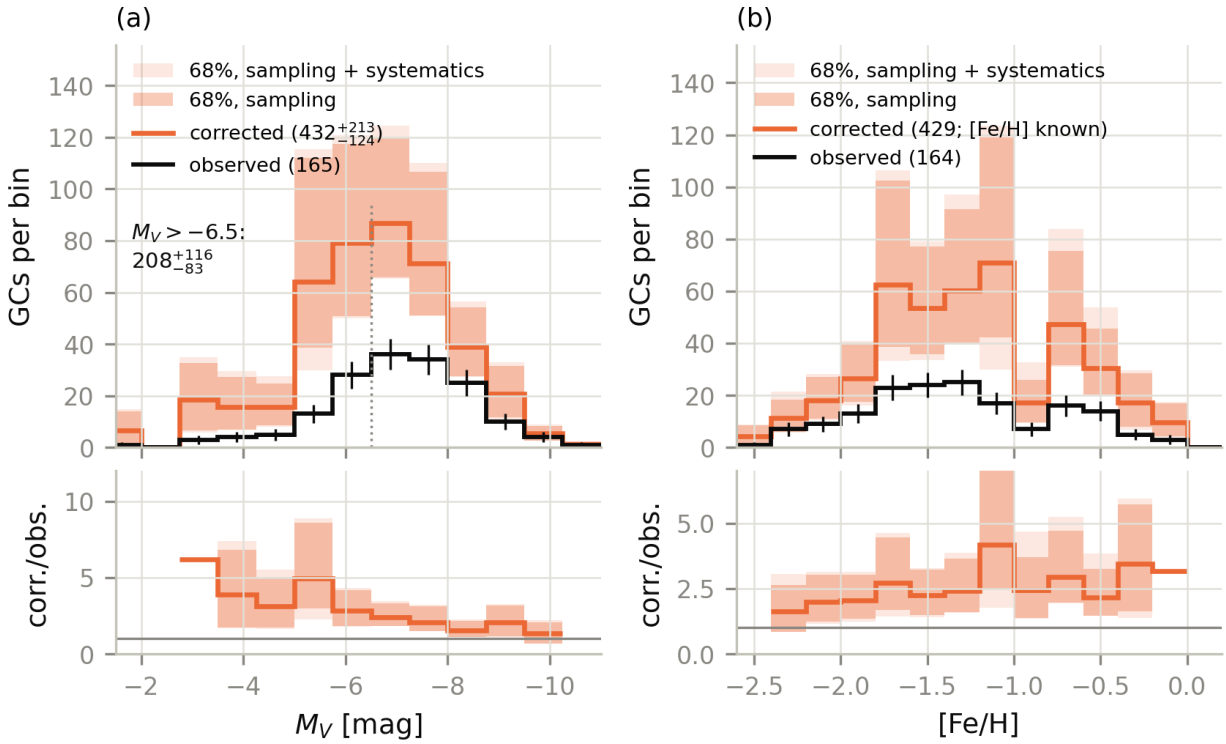


Figure 11: The final model: corrected M_V (a) and [Fe/H] (b) distributions with the continuous $C(V)$, 3D-map extinction (high bracket) and own-orbit $1/q$ weights. Dark band: 68% cluster bootstrap (sampling); light band: sampling and the quantified systematics in quadrature. The systematic part is evaluated in each bin from the per-cluster weights of every alternative in Table 14 (weighting, tail shape, direction-dependent C, map bracket): for each component, the largest downward and upward change of the bin count, then quadrature over components. The legend in (a) gives the total and the faint count with the same construction (Eq. 4); (b) omits the one cluster without [Fe/H]. Lower panels: corrected/observed. The bootstrap band is calibrated in mocks for the total and faint counts only; in bins with a few observed clusters it is coarse.

Component	down	up
Sampling (cluster bootstrap, 321–628)	−0.30	+0.37
Weighting (393–432)	−0.10	0
Faint-tail shape (424–457)	−0.02	+0.06
Direction-dependent C (380–485)	−0.13	+0.12
Map bracket (432–461)	0	+0.07
Quantified systematics, in quadrature	−0.16	+0.14
Total, in quadrature	−0.34	+0.40

The total gives factors of $e^{-0.34} = 0.71$ and $e^{0.40} = 1.49$, i.e. 308–645 around 432. The sampling part is calibrated: in mocks matched to the observed sample, the 68% bootstrap intervals cover the truth in 63–79% of realisations (Sec. 5.9). The systematic ranges are spreads between model choices, not Gaussian errors. Adding the systematic total linearly to the sampling error instead (−0.46, +0.52) gives 275–725; adding every component linearly (−0.54, +0.61) gives 250–795. For the faint count the same construction gives systematics of −0.29 and +0.16, larger than for the total: the three-region C (−0.24) and the $1/V_{\max}$ weighting (−0.16) both lower it, so the faint count is 208^{+116}_{-83} (125–325).

What is not in the error bar. Three effects are not quantified. If the brightest clusters, which anchor C, are themselves incomplete, every weight rises; clusters on orbits that are never detectable

are counted by none of our weightings. Both can only increase N. Errors in the Galactic potential, the orbits and the phase-mixing assumption have unknown sign. Equation 4 is therefore best read as about 430, most likely between 310 and 650, with the unquantified effects pointing upwards.

7 Limitations and next steps

1. **Unseen orbit families.** No weighting counts orbit families with no detected member unless an orbit population is assumed. The data favour at most a mild trend of orbit size with luminosity (Sec. 5.9); a stronger one would make all four weightings low for faint clusters.

2. **Extinction.** The 3D-map treatment rests on fixed conversions (in particular $A_V = 8.8\,A_{K_s}$ for Marshall, uncertain in the inner Galaxy) and on SFD beyond the maps' reach. Map uncertainties and alternative extinction laws are not propagated beyond the low/high brackets; mocks with map-based rather than analytic dust would test the full chain.

3. **Anchor.** All absolute numbers assume the brightest-V clusters are complete. Every weight scales as $1/C_{\mathrm{anchor}}$; an independent census of the brightest GCs would test it.

4. **Potential, phase mixing and orbit errors.** The intervals cover sampling noise only. An ensemble of perturbed phase-space coordinates and alternative potentials would give the systematic spread. The u test fails mildly for clusters outside the inner window ($p = 0.01$–0.025) under every completeness model (Sec. 5.10); groups that are not phase-mixed, such as the clusters of the Sagittarius dwarf, which all sit near the inner parts of their orbits, are one candidate.

5. **Mock realism.** The calibrated coverage of Sec. 5.9 holds for mocks whose true completeness is the fitted one and whose population is quasi-spherical and isotropic; the misspecified controls show biases of up to -0.35 in ln for faint clusters.

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